

# Does an NHL Teams available Cap Space predict Win percentage?

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Research Question: Does available cap space predict an NHL team's win percentage?

Null Hypothesis: There is no correlation between an NHL team's win% and how much cap space they have available.

Data Source: Spotrac, a website that tracks NHL teams available cap, Total team payroll, Win percentage, and other statistics that we didn't use for our project. For our project we specifically looked at the 2024-2025 season.

$$\text{WIN\%}(\text{excluding ties}) = B_0 + B_1 * (\text{Cap Space Available}) + B_2 * (\text{Total Team Payroll})$$

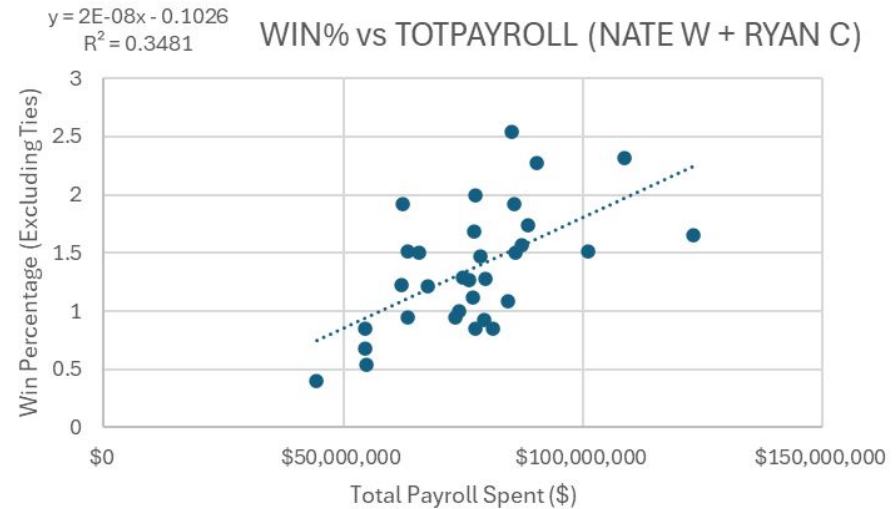


B1(Cap Space Available) =  $-6.81811\text{E-}09$ , p-value = 0.619347181  
B2(Total Team Payroll) =  $1.72822\text{E-}08$ , p-value of 0.007173717

The p-value for Cap Space Available being about 0.6193 tells us that there is a 61.93% chance that these results occurred due to random variation.

The p-value for Total Payroll Spent on players being about .007 tells us that there is only a 0.7% chance that our results occurred randomly.

The graph on slide 1 makes it seem like there is a correlation between cap space available and win percentage, as the data points are fairly spread out and the line of best fit shows the less available cap space, or available spending, you have, the higher your win percentage. Although the graph shows this, the p-value we calculated for cap space available is very high. A p-value of higher than 0.05 means that the results of the data are not statistically significant.



Conclusion: Teams that spend more win more, and cap space fails to account for the actual difference in the amount that teams spent on their players in the 2024-2025 season. For this reason we chose to fail to reject the null hypothesis.